

# Channel Quantization for Efficient Dual Lattice Attacks on Learning with Errors

Anonymous

**Abstract**—Dual Attacks are promising methods to break post-quantum lattice structures. Their distinguishing stage compresses sieve inner products before one transform scores every candidate. However, two problems are open. First, provable dual attacks admit no compressor based on coding. Second, the transform group, tied to the modulus  $q$ , excludes characteristic two. Read as a modulo lattice channel, compression becomes lattice quantization. We prove a dual attack correct for every nested lattice quantizer, build index groups of characteristic two reaching each integer table size, and show that one step quantization dominates staged switching and likelihood ratio combining is optimal. Measurements give four orders of magnitude less compression time and  $\times 58$  less memory.

**Index Terms**—Communication channel, source coding, lattice codes, post-quantum cryptanalysis, lattice quantization, dual lattice attack.

## I. Introduction

THE Learning With Errors problem (LWE) of Regev [1] asks to recover a short secret  $\mathbf{s}$  from a public uniform matrix  $\mathbf{A}$  and a vector  $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q$ , in which  $q$  is the modulus and  $\mathbf{e}$  a short error. It carries the security of ML-KEM [2], [3] and ML-DSA [4], and therefore of a growing share of the key material that protects real communication channels; parameters for these schemes are set by the primal and the dual attack.

A dual attack [5], [6] splits the columns of  $\mathbf{A}$  into a guessed part and a part left to lattice reduction, and uses a sieve [7], [8] to produce short vectors of a dual lattice. Each such vector turns the guessed part into one noisy equation modulo  $q$ , and a score summing  $\cos(2\pi \cdot /q)$  over the list separates the correct guess from the wrong ones [9]–[11]. Guo and Johansson [12] observed that all candidate scores are values of one discrete Fourier transform, and MATZOV [13] added modulus switching [14] so that the transform runs over  $\mathbb{Z}_p^{n_f}$  with  $p \ll q$ , in which  $n_f$  is the number of guessed coordinates handled by the transform. Carrier *et al.* [15] replaced modulus switching by a lossy source code built from polar codes over  $\mathbb{Z}_q$  [16], which lowers the extra noise at equal table size. In parallel, Ducas and Pulles [17], [18] showed that the independence assumption behind the published analyses is not consistent, and Pouly and Shen [19] gave the first dual attack proved correct without statistical assumptions.

Each of these compressors is a lattice quantizer, and this is the point of view we adopt because mapping an inner product vector onto a coarser grid is what a receiver does before detection in a digital link, so the figures of merit are those of source coding, the normalized second moment and the covering radius, while the score is a detector for a modulo lattice additive noise channel [20]. Two questions raised in this line of work are still open.

**Problem 1.** Pouly and Shen [19] leave as an open problem the integration of a compressor into their provable framework and point at lattice codes as the variant most likely to admit a formal analysis. Qu and Xu [21] restate that problem and settle it for scalar modulus switching alone, at the price of one run per prime of a coprime family and a recombination of residues by the Chinese remainder theorem. Wang *et al.* [22] again list the coding-based compressor as open and ask whether it can shrink the search space from  $q^{n_f}$  to  $q^{n'}$  with  $n' \ll n_f$  while a rigorous guarantee is kept.

**Problem 2.** Carrier *et al.* [15] list in their concluding remarks the combination of modulus switching with their source code, because it would move the transform into characteristic two and make it much cheaper, and Ducas and Pulles [17] state that the group of the transform should be  $(\mathbb{Z}/2\mathbb{Z})^k$  and give a sparsification that reaches it but compresses nothing. Neither

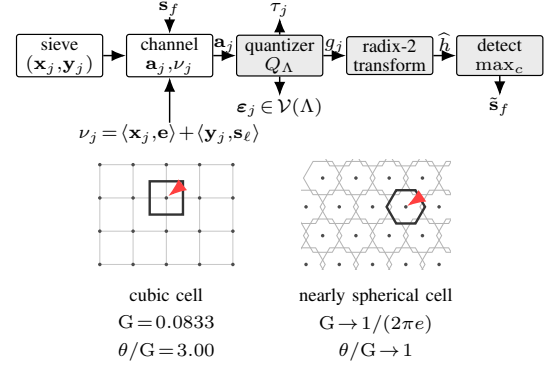


Fig. 1. Above, the dual attack as a communication link: a sieve produces short vectors  $(\mathbf{x}_j, \mathbf{y}_j)$ , each turning  $\mathbf{s}_f$  into one observation of a modulo- $q$  additive noise channel with gain  $\mathbf{a}_j$  and noise  $\nu_j$ , and the shaded blocks are the stages redesigned here, the quantizer  $Q_\Lambda$  that compresses  $\mathbf{a}_j$  into a group index  $g_j$  and leaves a residual  $\epsilon_j$ , the transform, and the detector, with the reliability  $\tau_j$  weighting the sample. Below, that compression in two dimensions, with black dots the quantizer points, thin outlines their Voronoi cells, the red dot a gain vector and the red arrow its residual: cubic cells have a worst case three times their average, nearly spherical cells bring the worst case down to the average.

construction is instantiated for LWE, and the coupling of the group to  $q$  also fixes the granularity of the table at  $\log_2 q \approx 11.7$  bits for ML-KEM, so a design that needs  $2^{95}$  entries has to pay for  $2^{105}$ .

## A. Contributions

*First*, we solve Problem 1. We formalize the compressor as a nested lattice quantizer  $\Lambda \supseteq q\mathbb{Z}^{n_f}$  and show that the candidate space collapses to  $\mathbb{Z}^{n_f}/q\Lambda^*$ , of cardinality  $N_G = q^{n_f}/\det \Lambda$ . Theorem 5 gives a correctness condition in which compression enters only through the covering radius  $\rho(\Lambda)$  and recovery only through  $\lambda_1(q\Lambda^*)$  as no independence assumption is used, the whole guessed block is returned by one run without residue recombination, and the search space  $N_G$  is free.

*Second*, we solve Problem 2 with quantizers whose index group is an abelian two group of prescribed order, so the transform is radix two and the table size is any power of two, a granularity of one bit instead of  $\log_2 q$ . Theorem 6 describes the group and the transform, and Theorem 7 shows that one nearest neighbor step in the composite lattice is never worse than switching the modulus and then applying a source code.

*Third*, we present NLQ, a distinguisher instantiating the framework with a direct sum of Leech lattices and small companions, an exact nearest neighbor decoder, a radix two transform and a likelihood ratio weighting of the scores, which Theorem 9 shows to maximize the deflection among all aggregations a single transform can evaluate. We implement it together with the truncation of [12], modulus switching [13], the polar source code of [15], its staged combination with switching [23] and the product of random codes, and compare them by complexity and by measurement.

Dual attacks in coding theory follow a parallel path [24], [25], and lattice quantization has a long history in communication theory, from the fast decoders of [26], [27] and the second moment tables of [28], [29] to nested lattice codes [20]; score aggregation with unequal reliabilities is diversity combining [30]. Li and Zheng [23] independently compose modulus switching with a polar source code, which Theorem 7 shows to be dominated by the single step quantizer.

TABLE I

BASE LATTICES USED BY ALLOCATE, WITH BLOCK DIMENSION  $n_b$ , INDEX  $\det\Lambda_b$  IN  $\mathbb{Z}^{n_b}$ , CONDUCTOR  $c_b$ , THE LEAST POWER OF TWO WITH  $c_b\mathbb{Z}^{n_b} \subseteq \Lambda_b$ , NORMALIZED SECOND MOMENT  $G$ , NORMALIZED SQUARED COVERING RADIUS  $\theta$ , THEIR RATIO, THE COST OF AN EXACT NEAREST NEIGHBOR DECODER, AND THE ALLOCATION WEIGHT  $\gamma_b = G(\det\Lambda_b)^{2/n_b}$ . THE LAST TWO ROWS ARE INTEGRAL REALIZATIONS OF THE LEECH LATTICE OF EQUAL  $G$  BUT DIFFERENT CONDUCTORS, SO THEY REACH DIFFERENT RATES.

| $\Lambda_b$     | $n_b$ | $\det\Lambda_b$ | $c_b$ | $G(\Lambda_b)$ | $\theta(\Lambda_b)$ | $\theta/G$ | ops | $\gamma_b$ |
|-----------------|-------|-----------------|-------|----------------|---------------------|------------|-----|------------|
| $\mathbb{Z}$    | 1     | 1               | 1     | 0.08333        | 0.2500              | 3.00       | 3   | 0.0833     |
| $D_4$           | 4     | 2               | 2     | 0.07664        | 0.1768              | 2.31       | 24  | 0.1084     |
| $D_7$           | 7     | 2               | 2     | 0.07567        | 0.2051              | 2.71       | 42  | 0.0922     |
| $\Lambda_{E_8}$ | 8     | $2^4$           | 2     | 0.07168        | 0.1250              | 1.74       | 104 | 0.1434     |
| $\Lambda_{24}$  | 24    | $2^{36}$        | 8     | 0.06578        | 0.08333             | 1.27       | 331 | 0.5263     |
| $\Lambda'_{24}$ | 24    | $2^{24}$        | 4     | 0.06578        | 0.08333             | 1.27       | 379 | 0.2631     |

## II. Preliminaries

Here, we address the background components we focus in this work. For the proofs of statements and further details, refer to appendix of full version: <https://anonymous.4open.science/r/work-37-paper>.

Vectors are columns in bold, and  $\mathbb{Z}_q$  is identified with the centered representatives in  $(-q/2, q/2]$ . The centered binomial distribution  $\chi_\eta$ , which is the secret and error distribution of ML-KEM, has variance  $\sigma^2 = \eta/2$  and characteristic function  $\varphi_\eta(t) = \mathbb{E}[\cos(tX)] = \cos^{2\eta}(t/2)$ .

A lattice  $\Lambda \subset \mathbb{R}^n$  has determinant  $\det\Lambda$ , Voronoi cell  $\mathcal{V}(\Lambda)$ , minimum distance  $\lambda_1(\Lambda)$ , covering radius  $\rho(\Lambda) = \max_{\mathbf{x}} \|\mathbf{x} - Q_\Lambda(\mathbf{x})\|$  with  $Q_\Lambda$  the nearest-neighbor map, and dual  $\Lambda^*$ . Its normalized second moment, which measures average distortion, is  $G(\Lambda) = (n(\det\Lambda)^{1+2/n})^{-1} \int_{\mathcal{V}(\Lambda)} \|\mathbf{x}\|^2 d\mathbf{x}$ ; it is invariant under scaling and rotation, equals  $1/12$  for  $\mathbb{Z}$  and exceeds  $1/(2\pi e) \approx 0.0586$  in every dimension [20], [28], and  $\theta(\Lambda) = \rho(\Lambda)^2 / (n(\det\Lambda)^{2/n})$  is its worst-case counterpart. Two base lattices carry the construction as  $\Lambda_{E_8} = \{\mathbf{x} \in \mathbb{Z}^8 : \mathbf{x} \bmod 2 \in \mathcal{H}_8\}$ , with  $\mathcal{H}_8$  the extended Hamming code, and the integral Leech lattice  $\Lambda_{24} = 2\Lambda_B \cup (2\Lambda_B + \mathbf{u})$  with  $\Lambda_B = \{\mathbf{y} \in \mathbb{Z}^{24} : \mathbf{y} \bmod 2 \in \mathcal{G}_{24}, \sum_i y_i \equiv 0 \bmod 4\}$ ,  $\mathcal{G}_{24}$  the extended binary Golay code and  $\mathbf{u} = (-3, 1, \dots, 1)$ ; rotating the latter by the matrix with rows  $\mathbf{e}_{2k-1} \pm \mathbf{e}_{2k}$  and halving gives a second integral copy  $\Lambda'_{24}$  of determinant  $2^{24}$ .

An LWE instance is  $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$  with  $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}$  and entries of  $\mathbf{s}, \mathbf{e}$  from  $\chi_\eta$ . Following [10], [13] we split  $n = n_e + n_f + n_\ell$ , write  $\mathbf{A} = [\mathbf{A}_e | \mathbf{A}_f | \mathbf{A}_\ell]$  and  $\mathbf{s} = (\mathbf{s}_e, \mathbf{s}_f, \mathbf{s}_\ell)$ , and enumerate the first block, recover the second by a transform and absorb the third by lattice reduction. A sieve on  $\{(\mathbf{x}, \mathbf{y}) \in \mathbb{Z}^m \times \mathbb{Z}^{n_\ell} : \mathbf{A}_\ell^\top \mathbf{x} \equiv \mathbf{y} \bmod q\}$  returns  $N$  short pairs  $\mathbf{w}_j = (\mathbf{x}_j, \mathbf{y}_j)$  with  $\langle \mathbf{x}_j, \mathbf{b} - \mathbf{A}_e \mathbf{s}_e \rangle - \langle \mathbf{a}_j, \mathbf{s}_f \rangle \equiv \nu_j \pmod{q}$ , (1) with gain vector  $\mathbf{a}_j = \mathbf{A}_f^\top \mathbf{x}_j \in \mathbb{Z}_q^{n_f}$  and noise  $\nu_j = \langle \mathbf{x}_j, \mathbf{e} \rangle + \langle \mathbf{y}_j, \mathbf{s}_\ell \rangle$ .

## III. The Dual Attack Channel

Equation (1) is a memoryless channel over  $\mathbb{Z}_q$  with input  $\mathbf{s}_f$ , gain vector  $\mathbf{a}_j$  and additive noise  $\nu_j$ , as drawn in Fig. 1. Since the  $\mathbf{a}_j$  are essentially uniform, scoring all  $q^{n_f}$  candidates is a transform over  $\mathbb{Z}_q^{n_f}$ , far too large; compression replaces  $\mathbf{a}_j$  by a point of a coarser lattice, a lossy source coding step in front of the detector.

**Definition 1** (Quantizer lattice and index group). A quantizer lattice is a full-rank  $\Lambda \subset \mathbb{R}^{n_f}$  with  $q\mathbb{Z}^{n_f} \subseteq \Lambda$ . Its index group is the finite abelian group  $G_\Lambda = \Lambda / q\mathbb{Z}^{n_f}$  of order  $N_G = q^{n_f} / \det\Lambda$ , its candidate lattice is  $\Lambda_o = q\Lambda^* \subseteq \mathbb{Z}^{n_f}$ , and its residual on input  $\mathbf{a}$  is  $\varepsilon = \mathbf{a} - Q_\Lambda(\mathbf{a}) \in \mathcal{V}(\Lambda)$ .

With  $\lambda_j = Q_\Lambda(\mathbf{a}_j)$  and  $t_j = \langle \mathbf{x}_j, \mathbf{b} - \mathbf{A}_e \tilde{\mathbf{s}}_e \rangle$ , the compressed score of a candidate  $\tilde{\mathbf{s}} \in \mathbb{Z}^{n_f}$  with weights  $w_j \geq 0$  is

$$F_{\tilde{\mathbf{s}}}(\tilde{\mathbf{s}}) = \sum_{j=1}^N w_j \cos\left(\frac{2\pi}{q}(t_j - \langle \lambda_j, \tilde{\mathbf{s}} \rangle)\right). \quad (2)$$

## Algorithm 1 Provable coded dual attack

**Require:**  $(\mathbf{A}, \mathbf{b})$  with  $\mathbf{A} = [\mathbf{A}_g | \mathbf{A}_d]$ , quantizer lattice  $\Lambda$ , width  $s$ , count  $N$ , bound  $B \geq \|\mathbf{s}_g\|$   
**Ensure:** the guessed block  $\mathbf{s}_g$   
1: sample  $\mathbf{w}_1, \dots, \mathbf{w}_N$  independently from  $D_{L_q^\perp}(\mathbf{A}_d, q, s)$   
2:  $h \leftarrow 0$  on  $G_\Lambda$   
3: **for**  $j = 1$  to  $N$  **do**  
4:  $\mathbf{a}_j \leftarrow \mathbf{A}_g^\top \mathbf{w}_j \bmod q$  centered,  $\lambda_j \leftarrow Q_\Lambda(\mathbf{a}_j)$   
5:  $h(g_j) \leftarrow h(g_j) + e^{2\pi i \langle \mathbf{w}_j, \mathbf{b} \rangle / q}$ ,  $g_j = \lambda_j + q\mathbb{Z}^{n_f}$   
6: **end for**  
7:  $\tilde{h} \leftarrow$  Fourier transform of  $h$  over  $G_\Lambda$   
8:  $\mathcal{C} \leftarrow$  cosets of  $\Lambda_o$  with a representative of norm  $\leq B$   
9: **return** the shortest representative of  $\arg\max_{c \in \mathcal{C}} \text{Re} \tilde{h}(c)$

Lemma 2 identifies the candidate space and the transform.

**Lemma 2.** *The score (2) depends on  $\tilde{\mathbf{s}}$  only through its coset in  $\mathbb{Z}^{n_f} / \Lambda_o$ , the index  $[\mathbb{Z}^{n_f} : \Lambda_o]$  equals  $N_G$ , and  $\tilde{\mathbf{s}} \mapsto \chi_{\tilde{\mathbf{s}}}$  with  $\chi_{\tilde{\mathbf{s}}}(\lambda) = \exp(2\pi i \langle \lambda, \tilde{\mathbf{s}} \rangle / q)$  is an isomorphism from  $\mathbb{Z}^{n_f} / \Lambda_o$  onto the character group of  $G_\Lambda$ . All  $N_G$  distinct scores are therefore the real parts of one Fourier transform over  $G_\Lambda$ .*

Lemma 2 makes the design space explicit since a quantizer lattice is graded by the transform size  $N_G$ , the average distortion  $n_f G(\Lambda) (\det\Lambda)^{2/n_f}$ , the worst-case distortion  $\rho(\Lambda)^2$  and the minimum distance  $\lambda_1(\Lambda_o)$ , which decides whether a coset determines the secret. The trade-off is quantitative as with  $\tau_j$  the bias of sample  $j$ , the correct score grows as  $N\bar{\tau}$  while a wrong one has deviation  $\sqrt{N/2}$  by Lemma 3, so about  $2\ln(N_G)/\bar{\tau}^2$  samples separate them and a good compressor shrinks  $N_G$  without driving  $\bar{\tau}$  down.

Table II gives the lattice of each published compressor as truncation [12] and modulus switching [13] take  $\Lambda = (q/p)\mathbb{Z}^{n_f}$ , the source code of [15] with generator  $\mathbf{G}$  takes the Construction-A lattice of  $\langle \mathbf{G} \rangle$ , a product of random codes gives a direct sum of such lattices, and the staged design of [23] takes  $(q/p)\Lambda'$  with  $\Lambda'$  a Construction-A lattice over  $\mathbb{Z}_p$ .

The second moment of the score over the candidate set is fixed by Parseval and needs no assumption, which is where our analysis departs from the independence heuristic criticized in [17].

**Lemma 3.** *Let  $h(g) = \sum_{j: g_j = g} w_j e^{2\pi i t_j / q}$  be the accumulator, in which  $g_j$  is the index of  $\lambda_j$ . Then the average of  $F_{\tilde{\mathbf{w}}}$  over the  $N_G$  cosets is  $\text{Re} h(0)$  and*

$$\frac{1}{N_G} \sum_{\tilde{\mathbf{s}}} F_{\tilde{\mathbf{w}}}(\tilde{\mathbf{s}})^2 = \frac{1}{2} \left( \sum_g |h(g)|^2 + \text{Re} \sum_g h(g) h(-g) \right),$$

*which equals  $\frac{1}{2} \sum_j w_j^2$  when the indices  $g_j$  are pairwise distinct and no two of them are inverse to each other.*

## IV. A Provable Coded Dual Attack

We now place an arbitrary quantizer lattice inside the provable framework of [19], which replaces the sieve by discrete Gaussian sampling on the dual lattice and the heuristic score model by the pointwise approximation of [6]. Write  $\rho_{1/s}(\mathbf{x}) = \exp(-\pi s^2 \|\mathbf{x}\|^2)$ , let  $L_q(\mathbf{M})$  be the  $q$ -ary lattice generated by the columns of  $\mathbf{M}$  and  $q\mathbb{Z}^m$ , and  $D_{L, \sigma}$  the discrete Gaussian on  $L$  of width  $\sigma$ .

Algorithm 1 takes the instance, a quantizer lattice, a width  $s$  and a bound  $B$  on the guessed block; it draws  $N$  dual vectors from a discrete Gaussian, which is the step that replaces the sieve and makes the analysis unconditional, then for each sample forms the gain vector  $\mathbf{a}_j$ , quantizes it to the nearest point  $\lambda_j$  of  $\Lambda$  and adds the phase  $e^{2\pi i \langle \mathbf{w}_j, \mathbf{b} \rangle / q}$  to the accumulator entry indexed by the coset  $g_j$ ; after one transform, which yields all  $N_G$  scores, it keeps only the cosets holding a vector of norm at most  $B$ , because the secret is short, and returns the shortest representative of the winner. The next lemma replaces each distributional statement about rounding by a geometric one.

TABLE II

COMPRESSORS AS INSTANCES OF DEFINITION 1, WITH QUANTIZER LATTICE  $\Lambda$ , IN WHICH  $p$  IS THE SWITCHED MODULUS AND  $\langle \mathbf{G} \rangle$  THE CODE GENERATED BY  $\mathbf{G}$ ; INDEX GROUP, OF ORDER THE TABLE SIZE  $N_G$ ; GRANULARITY, THE SPACING OF ATTAINABLE TABLE SIZES; OPERATION COUNT AND WORKING MEMORY OF ONE COMPRESSION, WITH  $\mu(q) = \Theta(q \log q)$  AND  $n'$  THE LEAST POWER OF TWO AT LEAST  $n_f$ ; SMALLEST EFFECTIVE SECOND MOMENT ATTAINED AT  $n_f = 48$ ; RATIO OF WORST-CASE TO AVERAGE DISTORTION, WHICH A PROVABLE ANALYSIS NEEDS AND WHICH IS KNOWN ONLY FOR THE LATTICES OF KNOWN COVERING RADIUS; AND AVAILABILITY OF A PROOF FREE OF STATISTICAL ASSUMPTIONS AND OF THE OPTIMAL AGGREGATION OF THEOREM 9. SMALLER IS BETTER IN COLUMNS FOUR TO EIGHT.

| compressor                       | quantizer lattice $\Lambda$                                                      | index group $G_\Lambda$        | granularity                | compression ops             | memory                | $G_{\text{eff}}$ | $\theta/G$  | provable    | optimal w  |
|----------------------------------|----------------------------------------------------------------------------------|--------------------------------|----------------------------|-----------------------------|-----------------------|------------------|-------------|-------------|------------|
| Truncation [12]                  | $(q/p)\mathbb{Z}^{n_f}$                                                          | $\mathbb{Z}_p^{n_f}$           | $n_f \log_2 \frac{p+1}{p}$ | $\Theta(n_f)$               | $O(n_f)$              | 0.0833           | 3.00        | no          | no         |
| Modulus switching [13]           | $(q/p)\mathbb{Z}^{n_f}$                                                          | $\mathbb{Z}_p^{n_f}$           | $n_f \log_2 \frac{p+1}{p}$ | $\Theta(n_f)$               | $O(n_f)$              | 0.0833           | 3.00        | partly [21] | no         |
| Random product code              | $\bigoplus \{\mathbf{v} : \mathbf{v} \bmod q \in \langle \mathbf{G}_i \rangle\}$ | $\mathbb{Z}_q^{(n_f/n_0)k_0}$  | $\frac{n_f}{n_0} \log_2 q$ | $\Theta(n_f q^{k_0})$       | $\Theta(q^{k_0} n_0)$ | 0.0879           | n/a         | no          | no         |
| Polar source code [15]           | $\{\mathbf{v} : \mathbf{v} \bmod q \in \langle \mathbf{G} \rangle\}$             | $\mathbb{Z}_q^{k_f}$           | $\log_2 q \approx 11.7$    | $\Theta(n' \log n' \mu(q))$ | $\Theta(n' q)$        | 0.0654           | n/a         | no          | no         |
| Staged switching and coding [23] | $(q/p)\{\mathbf{v} : \mathbf{v} \bmod p \in \langle \mathbf{G} \rangle\}$        | $\mathbb{Z}_p^{k_f}$           | $\log_2 p$                 | $\Theta(n' \log n' \mu(p))$ | $\Theta(n' p)$        | 0.0677           | n/a         | no          | no         |
| <b>Our Work</b>                  | $\bigoplus_b (q/c_b m_b) \Lambda_b$                                              | $\prod_j \mathbb{Z}_{2^{a_j}}$ | 1                          | $\Theta(n_f)$               | $O(n_f)$              | <b>0.0654</b>    | <b>1.27</b> | <b>yes</b>  | <b>yes</b> |

TABLE III

BASE-TWO LOGARITHM OF THE COST OF THE DISTINGUISHING STAGE AT THE THREE ML-KEM PARAMETER SETS, IN THE GATE COUNT MODEL OF [31] APPLIED IDENTICALLY TO ALL DESIGNS, WITH THE SPLIT, THE SIEVE DIMENSION AND THE TABLE SIZE OPTIMIZED FOR EACH DESIGN UNDER ITS OWN GRANULARITY CONSTRAINT. THE COLUMNS GIVE THE TOTAL, THE COMPRESSION TERM, WHICH IS WHAT AN IMPLEMENTATION PAYS PER DUAL VECTOR AND DOMINATES ONCE THE SIEVE OUTPUT IS SHARED BETWEEN TARGETS, AND THE MEMORY. THE LEAST ENTRY OF EACH COLUMN IS IN BOLD.

| design                              | ML-KEM-512   |             |           | ML-KEM-768   |              |            | ML-KEM-1024  |              |            |
|-------------------------------------|--------------|-------------|-----------|--------------|--------------|------------|--------------|--------------|------------|
|                                     | tot.         | cmp.        | mem.      | tot.         | cmp.         | mem.       | tot.         | cmp.         | mem.       |
| Modulus switching [13]              | 149.7        | <b>94.0</b> | 100       | 217.0        | 135.0        | 171        | 294.9        | 181.9        | 249        |
| Random product code                 | 152.4        | 106.6       | 112       | 219.2        | 147.3        | 183        | 298.0        | 194.3        | <b>229</b> |
| Polar source code [15]              | 149.5        | 114.5       | 101       | 216.2        | 155.8        | 183        | 292.4        | 202.1        | 253        |
| Staged switching and coding [23]    | 149.4        | 110.2       | 103       | <b>215.5</b> | 152.2        | 169        | 292.5        | 197.7        | 241        |
| Our Work, $\Lambda_{E_8}$ and $D_k$ | 149.2        | 95.3        | <b>95</b> | 216.0        | <b>134.5</b> | <b>167</b> | 293.2        | <b>180.9</b> | 247        |
| Our Work, $\Lambda_{24}$            | <b>149.1</b> | 95.5        | 103       | 215.6        | 135.8        | <b>167</b> | <b>292.3</b> | 182.4        | 247        |

**Lemma 4.** For all  $\mathbf{a}$  and  $\mathbf{z}$ ,  $|\langle \mathbf{a} - Q_\Lambda(\mathbf{a}), \mathbf{z} \rangle| \leq \rho(\Lambda) \|\mathbf{z}\|$ . If  $\mathbf{a}$  is uniform on  $\mathbb{R}^{n_f}/q\mathbb{Z}^{n_f}$  and  $\mathbf{z}$  has independent entries of variance  $\sigma^2$ , then  $\mathbb{E}[\langle \mathbf{a} - Q_\Lambda(\mathbf{a}), \mathbf{z} \rangle^2] = \sigma^2 n_f G(\Lambda) (\det \Lambda)^{2/n_f}$ .

Theorem 5 states the correctness of the provable coded dual attack.

**Theorem 5.** Let  $q$  be a prime power,  $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$  of full rank with  $m \geq n$ ,  $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}$ ,  $\|\mathbf{s}_g\| \leq B$ , and let  $\Lambda$  be a quantizer lattice. Put  $\tau = s^{-1} \sqrt{m/2\pi}$  and assume  $\lambda_1(\Lambda_o) > 2B$ ,  $2B < q$ ,  $\lambda_1(L_q(\mathbf{A})) \geq \|\mathbf{e}\| + \tau$  and

$$\rho_{1/s}(\|\mathbf{e}\|) - \rho_{1/s}(\lambda_1(L_q(\mathbf{A})) - \|\mathbf{e}\| - \tau) > 2\delta + \frac{4\pi\rho(\Lambda)B}{q}. \quad (3)$$

Then Algorithm 1 outputs  $\mathbf{s}_g$  with probability at least  $1 - q^m 2^{-\Omega(N\delta^2)}$  over the sampling, in time  $\text{poly}(m, n)(N + N_G \log N_G)$  and space  $O(N_G + mn)$ .

Three features separate Theorem 5 from the provable variants that switch the modulus [21], [22] as compression contributes the single term  $4\pi\rho(\Lambda)B/q$ , a worst-case quantity free of any assumption on the residual; the whole block is returned by one run, because  $\Lambda_o$  separates short vectors as soon as  $\lambda_1(\Lambda_o) > 2B$ ; and the search space  $N_G$  is free by Definition 1 instead of a power of a small prime. What matters in (3) is  $\theta(\Lambda)$  and not  $G(\Lambda)$ , and the ratio  $\theta/G$  is 3 for the cubic lattice against 1.27 for  $\Lambda_{24}$ , so a good quantizer makes the worst case almost as favorable as the average, as Fig. 1 shows.

## V. Index Groups of Characteristic Two

Let  $\Lambda_1, \dots, \Lambda_B$  be base lattices with  $\Lambda_b \subseteq \mathbb{Z}^{n_b}$  and  $c_b \mathbb{Z}^{n_b} \subseteq \Lambda_b$  for a power of two  $c_b$ , called the conductor, and let  $m_b = 2^{r_b}$ . Set

$$\Lambda = \bigoplus_{b=1}^B \frac{q}{c_b m_b} \Lambda_b, \quad n_f = \sum_b n_b. \quad (4)$$

Theorem 6 describes the transform attached to (4).

**Theorem 6.** The lattice (4) is a quantizer lattice with  $\log_2 N_G = \sum_b (n_b \log_2(c_b m_b) - \log_2 \det \Lambda_b)$ , which is an integer and can be made equal to any prescribed nonnegative integer by an allocation over the base lattices of Table I. Its index group is  $G_\Lambda \cong \bigoplus_b \Lambda_b / (c_b m_b) \mathbb{Z}^{n_b}$ , an abelian two-group  $\prod_j \mathbb{Z}_{2^{a_j}}$  with  $\sum_j a_j = \log_2 N_G$  read off the Smith normal form of  $c_b m_b \mathbf{B}_b^{-1}$ , in which  $\mathbf{B}_b$  is a basis of  $\Lambda_b$ . All  $N_G$  scores follow from  $N_G \log_2 N_G$

complex additions and at most  $\frac{1}{2} N_G \log_2 N_G$  multiplications by roots of unity of two-power order, and from additions alone when every invariant factor equals two. If moreover  $\Lambda_b^* = c_b^{-1} \Lambda_b$ , then  $\Lambda_o = \bigoplus_b m_b \Lambda_b$  and  $\lambda_1(\Lambda_o) = \min_b m_b \lambda_1(\Lambda_b)$ .

The last condition holds for  $\mathbb{Z}$ ,  $\Lambda_{E_8}$  and  $\Lambda_{24}$ , so one routine both quantizes and returns the shortest representative of a coset. Theorem 7 compares one step with staged compression.

**Theorem 7.** Let  $\Lambda \subseteq \Lambda'$  be nested quantizer lattices. For all  $\mathbf{a}$ ,  $\|\mathbf{a} - Q_\Lambda(\mathbf{a})\| \leq \|\mathbf{a} - Q_{\Lambda'}(\mathbf{a})\|$ . If  $\mathbf{a}$  is uniform modulo  $q\mathbb{Z}^{n_f}$  and  $\Lambda'$  is fine enough that its points are equidistributed in the cells of  $\Lambda$ , the two mean squared errors are  $n_f G(\Lambda) (\det \Lambda)^{2/n_f}$  and that value plus  $n_f G(\Lambda') (\det \Lambda')^{2/n_f}$ . For  $\Lambda' = (q/p)\mathbb{Z}^{n_f}$  the excess is  $n_f q^2 / (12p^2)$ .

Composing modulus switching with a source code, as in [23], therefore adds the whole distortion of the scalar stage, whereas  $(q/p)\Lambda'$  is decoded in one step at no extra cost when  $\Lambda'$  has a nearest-neighbor decoder over the reals. Lemma 8 says how blocks should be chosen.

**Lemma 8.** For a direct sum  $\bigoplus_b \alpha_b \Lambda_b$  of fixed total volume the distortion is smallest when the per-dimension contribution  $G(\Lambda_b) (\alpha_b^{n_b} \det \Lambda_b)^{2/n_b}$  is the same for all blocks, and the resulting normalized second moment is the weighted geometric mean  $\prod_b G(\Lambda_b)^{n_b/n_f}$ .

Because the scalings are powers of two, the allocation is an integer program solved by dynamic programming over a table  $D[d, r]$  holding the least distortion of a partial design that fills  $d$  of the  $n_f$  coordinates with  $r$  of the  $R$  bits: starting from the empty design, each reachable state is extended by one block, which may be any base lattice at any scaling  $m = 2^i$ , adding  $n_b \log_2(c_b m) - \log_2 \det \Lambda_b$  bits and  $n_b \gamma_b / (c_b m)^2$  to the distortion, in which  $\gamma_b = G(\Lambda_b) (\det \Lambda_b)^{2/n_b}$  is the last invariant of Table I; the entry  $D[n_f, R]$  is then the design of least distortion at exactly that table size, and the table is filled in  $O(n_f R)$  steps. We call this procedure ALLOCATE.

## VI. The Nested Lattice Distinguisher

Algorithm 2 is our distinguisher. It uses ALLOCATE, the exact decoders of [26], [27], the radix two transform of Theorem 6, and a weight per sample that we now justify.

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**Algorithm 2** NLQ distinguisher
 

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**Require:**  $(\mathbf{A}, \mathbf{b})$ , split  $(n_e, n_f, n_\ell)$ , sieve output  $W$ , target  $R$   
**Ensure:** a candidate for  $(\mathbf{s}_e, \mathbf{s}_f)$   
 1:  $\{\Lambda_b, m_b\} \leftarrow \text{ALLOCATE}(n_f, R)$ ,  $\Lambda \leftarrow \bigoplus_b (q/c_b m_b) \Lambda_b$   
 2: **for**  $j = 1$  to  $N$  **do**  
 3:    $\mathbf{a}_j \leftarrow \mathbf{A}_f^\top \mathbf{x}_j$ ,  $\boldsymbol{\lambda}_j \leftarrow Q\Lambda(\mathbf{a}_j)$ ,  $\boldsymbol{\varepsilon}_j \leftarrow \mathbf{a}_j - \boldsymbol{\lambda}_j$   
 4:    $\tau_j \leftarrow \prod_i \varphi_\eta(\frac{2\pi \mathbf{x}_{j,i}}{q}) \prod_i \varphi_\eta(\frac{2\pi \mathbf{y}_{j,i}}{q}) \prod_i \varphi_\eta(\frac{2\pi \boldsymbol{\varepsilon}_{j,i}}{q})$   
 5: **end for**  
 6: **for** each guess  $\tilde{\mathbf{s}}_e$  in decreasing order of probability **do**  
 7:    $t_j \leftarrow \langle \mathbf{x}_j, \mathbf{b} - \mathbf{A}_e \tilde{\mathbf{s}}_e \rangle \bmod q$  for all  $j$   
 8:    $h(g) \leftarrow \sum_{j: g_j = g} \tau_j e^{2\pi i t_j / q}$ ,  $\hat{h} \leftarrow$  radix two transform  
 9:   **if**  $\max_c \text{Re} \hat{h}(c) \geq T$  **return**  $\tilde{\mathbf{s}}_e$  and the shortest representative of  $\arg \max_c$   
 10: **end for**

---

Algorithm 2 first calls ALLOCATE to fix the blocks and thereby the quantizer lattice, then makes one pass over the sieve output in which each short vector is turned into its gain vector, quantized, and given the reliability  $\tau_j$  obtained by evaluating  $\varphi_\eta$  at the entries of  $\mathbf{x}_j$ ,  $\mathbf{y}_j$  and of the residual, which is the expected cosine this sample contributes; the enumerated block is then swept in decreasing order of likelihood, and for each guess the accumulator is filled with the weighted phases and transformed once, the algorithm stopping as soon as a coset passes the threshold  $T$ . Lines 2 to 5 do not depend on the guess and run once.

Recall that  $\tau_j = \mathbb{E}[\cos(2\pi(\nu_j + \langle \boldsymbol{\varepsilon}_j, \mathbf{s}_f \rangle)/q)]$  is the bias of sample  $j$ , and let the deflection of a weight vector be  $\Delta(\mathbf{w}) = \mathbb{E}[F_{\mathbf{w}}(\mathbf{s}_f)] / \sqrt{\text{Var}[F_{\mathbf{w}}(\tilde{\mathbf{s}})]}$  with  $\tilde{\mathbf{s}}$  a uniform wrong candidate. Theorem 9 identifies the best weights.

**Theorem 9.** *With the variance evaluated by Lemma 3,  $\Delta(\mathbf{w}) = \sqrt{2} \langle \mathbf{w}, \boldsymbol{\tau} \rangle / \|\mathbf{w}\|_2$  is maximized exactly at  $\mathbf{w} \propto \boldsymbol{\tau}$ , where it equals  $\sqrt{2} \|\boldsymbol{\tau}\|_2$ . The gain over uniform weights is  $\sqrt{N} \|\boldsymbol{\tau}\|_2 / \|\boldsymbol{\tau}\|_1 \geq 1$ , with equality if and only if all  $\tau_j$  are equal, and the number of samples needed for a fixed deflection is divided by the square of that gain.*

Corollary 10 states the effect of enlarging the list.

**Corollary 10.** *Appending a sample to  $W$  never lowers the optimal deflection, so the whole sieve database can be used and no length threshold has to be tuned.*

If one transform is to produce all scores then only the coefficient  $w_j$ , and not a function of the phase, may depend on the sample, so Theorem 9 is optimal over the entire class of aggregations the transform admits. The weights are free, since they multiply accumulator entries and cost  $O(m + n_f)$  per sample. This answers the direction raised in [18], where unequal weights are suggested to improve the separation between correct and wrong candidates.

### A. Complexity

Let  $\mu(x) = \Theta(x \log x)$  be the cost of a length- $x$  transform, which is what [32] gives for a prime length, and  $n'$  the least power of two at least  $n_f$ . Proposition 11 gives the cost of the two procedures.

**Proposition 11.** *Per input vector, scalar modulus switching costs  $\Theta(n_f)$  operations and  $O(n_f)$  words; the polar source code of [15] costs  $\Theta(n' \log n' \mu(q))$  operations and  $\Theta(n' q)$  words; its staged combination with modulus switching [23] costs  $\Theta(n_f + n' \log n' \mu(p))$  operations and  $\Theta(n' p)$  words; a product of random  $[n_0, k_0]_q$  codes costs  $\Theta((n_f/n_0) q^{k_0} n_0)$  operations and  $\Theta(q^{k_0} n_0)$  words; and Algorithm 2 costs  $\Theta(n_f)$  operations and  $O(n_f)$  words, with 331 operations per 24 dimensions for  $\Lambda_{24}$  and 104 per 8 dimensions for  $\Lambda_{E_8}$ . The transform needs  $\Theta(N_G \log_2 N_G)$  additions and no length- $q$  subtransform for Algorithm 2, against  $\Theta(N_G \log_2 N_G)$  operations with a constant proportional to  $\mu(q)/(q \log_2 q)$  for a transform over  $\mathbb{Z}_q^{k_f}$ , and  $\Theta(N_G)$  words of accumulator in all cases.*

Table II collects the consequences. Compression is linear in  $n_f$  with a small constant and needs no table of size  $q$ , so it is cheaper than the polar decoder by a factor  $n' \log n' \mu(q)/n_f$ , which passes  $10^5$  at ML-KEM parameters; the transform is radix two, so its

constant is the smallest available; and the attainable  $\log_2 N_G$  form the whole integer range, so a design never overshoots by more than one bit, while a transform over  $\mathbb{Z}_q^{k_f}$  overshoots by  $\log_2 q/2 \approx 5.8$  bits on average, a factor of about 56 in both time and memory.

### VII. Experimental Evaluation

All of our practical findings are reproducible and available at the repository of <https://anonymous.4open.science/r/work-37>. We conducted our experiments on an Apple MacBook Air (2022) with the M2 chip, A15 Bionic (64-bit ARM-based), an 8-core CPU featuring a base clock speed of 3.49 GHz, and 16GB of memory, Tahoe OS 26.2 build 25C56. Our source code reproduces the decoding distances published in [15] to within 3.3 per cent, and our reduction model reproduces the dual vector lengths published there to within 6.6 per cent.

**Distortion.** Fig. 2(a) and (b) give the effective second moment against the rate. Our construction attains 0.0658, the least value known for a lattice quantizer in dimension 24 [29], at every half-integer rate, because the two realizations of Table I have conductors 8 and 4; between those points the allocation falls back on  $E_8$  and  $D_8$  blocks and the value rises to at most 0.079, still below the 0.083 of modulus switching, so among compressors decoded in linear time ours is best at every rate.

**Compression and transform.** Fig. 2(c) and (d) give the cost of the two procedures of Proposition 11. At  $n_f = 48$  the Leech quantizer takes  $149 \mu\text{s}$  per vector and the  $E_8$  quantizer  $2.5 \mu\text{s}$ , against 87.6 ms for the polar source code, 5.8 ms for the staged variant and 2.5 ms for the product of random codes, ratios of 586 and 34855 against the strongest competitor; the working memory of one pass follows the same pattern, 81 MB and 0.9 MB against 4720 MB. At a table of about  $2^{20}$  entries the two-group transform needs 40 operations per entry against 96 over  $\mathbb{Z}_p^{n_f}$  and 383 over  $\mathbb{Z}_q^{k_f}$ , and the spacing of  $\log_2 N_G$  is one bit here against 11.7 for  $\mathbb{Z}_q^{k_f}$  and 8 to 20 for  $\mathbb{Z}_p^{n_f}$ .

**Separation and sample complexity.** Fig. 2(e) and (f) present the distinguisher on a scaled instance with  $n = 44$ ,  $m = 56$ ,  $q = 3329$ ,  $\eta = 6$ ,  $n_f = 24$ , a reduced lattice of dimension 76 and 6000 dual vectors. Our design separates the correct score best and needs the fewest dual vectors,  $1.4 \times 10^4$  against  $2.3 \times 10^4$  for modulus switching,  $2.8 \times 10^4$  for the staged design and  $3.2 \times 10^4$  for the polar source code at a table of the same order, and the weights of Theorem 9 lower every count further, by a factor between 1.10 and 3.48 that grows with the spread of the sample reliabilities.

**Complete recovery.** On an instance with  $n = 34$ ,  $m = 46$ ,  $q = 257$ ,  $n_f = 8$  and  $N = 8000$  dual vectors all designs recover  $\mathbf{s}_f$ ; the stage takes 4 ms of compression, 45 ms of transform and a peak of 70 MB for ours, against 4.0 s, 4 ms and 840 MB for the polar source code, whose granularity forces a 16-bit table where ours places a 20-bit one.

**Cost at standardized parameters.** Table III applies one cost model, described in the code and identical for all designs, to the three ML-KEM parameter sets, each entry being the best configuration a design reaches when the split, the sieve dimension and the table size are optimized under its own granularity constraint. Our design gives the lowest total at two of the three sets and comes within 0.1 bits of the best at the third, and its compression term is the lowest of the designs that reach the distortion of a source code; the totals stay close because producing short vectors dominates them, which is what makes the compression term the quantity an implementation feels. Fig. 2(g) gives the same objective as a surface.

### VIII. Conclusion

We formalized the distinguishing stage of the dual attack as a quantized communication channel, which gives a provable coded dual attack, a transform in characteristic two decoupled from the modulus, and a compression stage faster and smaller than that of all previous designs. Carrying the same treatment

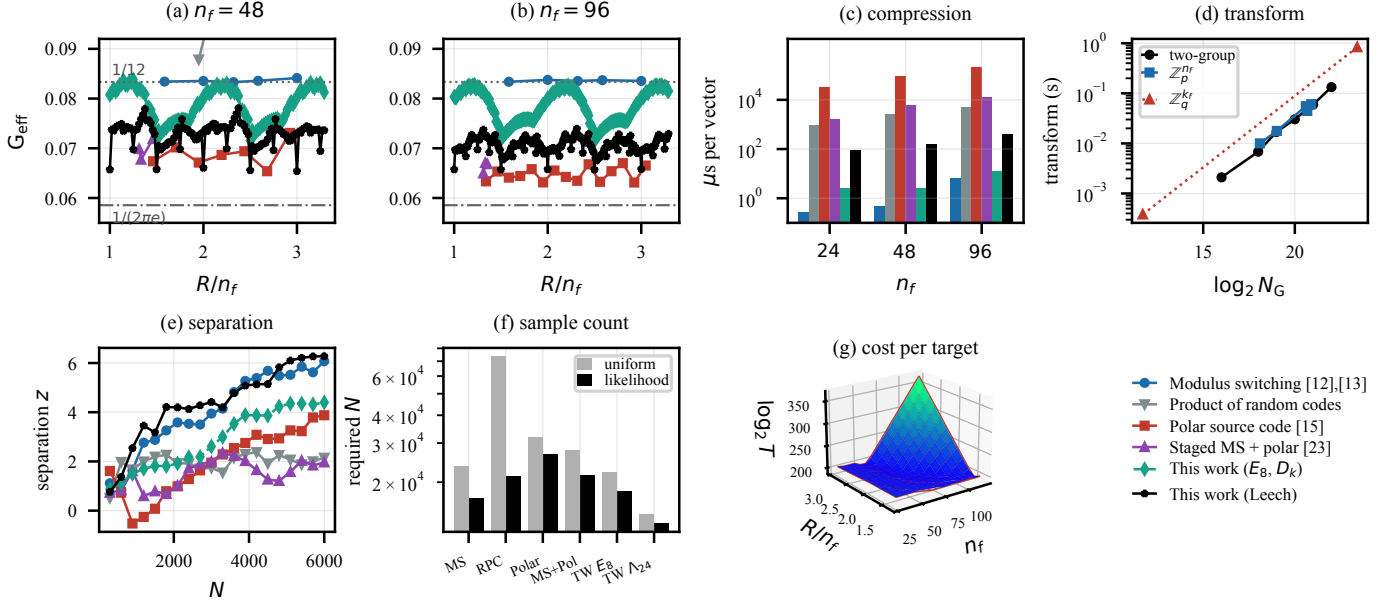


Fig. 2. Compression quality, cost and distinguishing behavior, all measured in one code base on one core. Panels (a) and (b) give the effective second moment  $G_{\text{eff}} = \mathbb{E} \|\varepsilon\|^2 / (n_f q^2 2^{-2R/n_f})$  against the rate  $R/n_f$  in bits per dimension on 800 uniform inputs with  $q = 3329$ , a marker existing only where a design can place a table and the horizontal lines being  $1/12$  and  $1/(2\pi e)$ . Panels (c) and (d) give the time to compress one gain vector and the time of one transform against the table size. Panel (e) gives the separation  $z = (F(\mathbf{s}_f) - \bar{F})/\sigma_F$  between the correct score and 4000 uniform candidates on an instance with  $n = 44$ ,  $m = 56$ ,  $q = 3329$ ,  $\eta = 6$ ,  $n_f = 24$ , panel (f) the dual vectors each design needs for the correct score to pass its largest wrong score, with uniform weights in gray and those of Theorem 9 in black, and panel (g) the cost per target for ML-KEM-768 with the sieve output shared, the red mesh being the polar source code and the filled surface ours.

to the nearest neighbor filter inside the sieve, and to the multistage handling of false positives can be considered as a future work.

## Appendix

This appendix collects the complete proofs of the statements of the main body. Throughout,  $\Lambda \subset \mathbb{R}^{n_f}$  is a quantizer lattice in the sense of Definition 1, that is a full-rank lattice with  $q\mathbb{Z}^{n_f} \subseteq \Lambda$ ,  $Q_\Lambda$  is its nearest-neighbor map,  $\mathcal{V}(\Lambda)$  its Voronoi cell,  $G_\Lambda = \Lambda/q\mathbb{Z}^{n_f}$  its index group,  $N_G = |G_\Lambda|$ , and  $\Lambda_o = q\Lambda^*$  its candidate lattice. For a finite abelian group  $G$  we write  $\hat{G}$  for its character group, and we use the character sum identity  $\sum_{\chi \in \hat{G}} \chi(g) = |G| \cdot \mathbf{1}[g=0]$ .

### A. Proof of Lemma 2

Fix  $\tilde{\mathbf{s}} \in \mathbb{Z}^{n_f}$  and consider the map  $\chi_{\tilde{\mathbf{s}}} : \Lambda \rightarrow \mathbb{C}^\times$  defined by  $\chi_{\tilde{\mathbf{s}}}(\boldsymbol{\lambda}) = \exp(2\pi i \langle \boldsymbol{\lambda}, \tilde{\mathbf{s}} \rangle / q)$ . It is a group homomorphism because the pairing is bilinear. It factors through  $G_\Lambda$ : for  $\mathbf{z} \in \mathbb{Z}^{n_f}$  we have  $\langle q\mathbf{z}, \tilde{\mathbf{s}} \rangle / q = \langle \mathbf{z}, \tilde{\mathbf{s}} \rangle \in \mathbb{Z}$ , so  $\chi_{\tilde{\mathbf{s}}}(q\mathbf{z}) = 1$  and  $q\mathbb{Z}^{n_f} \subseteq \ker \chi_{\tilde{\mathbf{s}}}$ . Hence  $\chi_{\tilde{\mathbf{s}}} \in \hat{G}_\Lambda$ , and  $\tilde{\mathbf{s}} \mapsto \chi_{\tilde{\mathbf{s}}}$  is a homomorphism  $\phi : \mathbb{Z}^{n_f} \rightarrow \hat{G}_\Lambda$ .

We compute  $\ker \phi$ . One has  $\chi_{\tilde{\mathbf{s}}} \equiv 1$  if and only if  $\langle \boldsymbol{\lambda}, \tilde{\mathbf{s}} \rangle \in q\mathbb{Z}$  for every  $\boldsymbol{\lambda} \in \Lambda$ , that is if and only if  $\langle \boldsymbol{\lambda}, \tilde{\mathbf{s}}/q \rangle \in \mathbb{Z}$  for every  $\boldsymbol{\lambda} \in \Lambda$ , that is if and only if  $\tilde{\mathbf{s}}/q \in \Lambda^*$ , that is  $\tilde{\mathbf{s}} \in q\Lambda^* = \Lambda_o$ . Moreover  $q\mathbb{Z}^{n_f} \subseteq \Lambda$  implies  $\Lambda^* \subseteq (q\mathbb{Z}^{n_f})^* = q^{-1}\mathbb{Z}^{n_f}$ , hence  $\Lambda_o \subseteq \mathbb{Z}^{n_f}$ , so  $\Lambda_o$  is a full-rank sublattice of  $\mathbb{Z}^{n_f}$  and the quotient  $\mathbb{Z}^{n_f}/\Lambda_o$  is finite of order  $\det(\Lambda_o) = q^{n_f} \det(\Lambda^*) = q^{n_f}/\det \Lambda = N_G$ .

On the other side,  $[\Lambda : q\mathbb{Z}^{n_f}] = \det(q\mathbb{Z}^{n_f})/\det(\Lambda) = q^{n_f}/\det \Lambda = N_G$ , so  $|G_\Lambda| = |\hat{G}_\Lambda| = N_G$ . Thus  $\phi$  induces an injective homomorphism from  $\mathbb{Z}^{n_f}/\Lambda_o$ , a group of order  $N_G$ , into  $\hat{G}_\Lambda$ , a group of the same order; an injective map between finite sets of equal cardinality is a bijection, so the induced map is an isomorphism.

Finally, writing  $h(g) = \sum_{j: g_j = g} w_j e^{2\pi i t_j / q}$  for  $g \in G_\Lambda$ , in which  $g_j = \boldsymbol{\lambda}_j + q\mathbb{Z}^{n_f}$ ,

$$\begin{aligned} F_{\mathbf{w}}(\tilde{\mathbf{s}}) &= \text{Re} \sum_j w_j e^{2\pi i t_j / q} \overline{\chi_{\tilde{\mathbf{s}}}(\boldsymbol{\lambda}_j)} \\ &= \text{Re} \sum_{g \in G_\Lambda} h(g) \overline{\chi_{\tilde{\mathbf{s}}}(g)} = \text{Re} \hat{h}(\chi_{\tilde{\mathbf{s}}}), \end{aligned}$$

so the score depends on  $\tilde{\mathbf{s}}$  only through  $\chi_{\tilde{\mathbf{s}}}$ , hence only through the coset of  $\tilde{\mathbf{s}}$  modulo  $\Lambda_o$ , and the family of all  $N_G$  scores is the real part of the Fourier transform of  $h$  over  $G_\Lambda$ . ■

### B. Proof of Lemma 3

Identify candidates with characters through Lemma 2 and write  $\hat{h}(\chi) = \sum_g h(g) \chi(g)$ . By that identity,

$$\frac{1}{N_G} \sum_{\chi} \hat{h}(\chi) = \sum_g h(g) \cdot \frac{1}{N_G} \sum_{\chi} \chi(g) = h(0),$$

and taking real parts gives the first claim. For the second,  $F = \text{Re} \hat{h} = (\hat{h} + \bar{\hat{h}})/2$ , so

$$F^2 = \frac{1}{4} (\hat{h}^2 + 2|\hat{h}|^2 + \bar{\hat{h}}^2) = \frac{1}{2} (|\hat{h}|^2 + \text{Re} \hat{h}^2).$$

Parseval for finite abelian groups gives  $\sum_{\chi} |\hat{h}(\chi)|^2 = N_G \sum_g |h(g)|^2$ . For the other term,

$$\sum_{\chi} \hat{h}(\chi)^2 = \sum_{g, g'} h(g) h(g') \sum_{\chi} \chi(g+g') = N_G \sum_g h(g) h(-g),$$

again by the same identity. Dividing by  $N_G$  and combining the two displays gives the stated identity. If the indices  $g_j$  are pairwise distinct then  $h$  is supported on  $\{g_1, \dots, g_N\}$  with  $|h(g_j)| = w_j$ , so  $\sum_g |h(g)|^2 = \sum_j w_j^2$ ; if in addition no two of the  $g_j$  are inverse to each other then  $h(g)h(-g) = 0$  for every  $g$ , and the second sum vanishes. ■

### C. Proof of Lemma 4

Write  $\varepsilon = \mathbf{a} - Q_\Lambda(\mathbf{a})$ . By the definition of the nearest-neighbor map,  $\varepsilon \in \mathcal{V}(\Lambda)$  and  $\|\varepsilon\| \leq \rho(\Lambda)$ , the covering radius being the largest distance from a point of  $\mathbb{R}^{n_f}$  to the lattice. Cauchy-Schwarz then gives  $|\langle \varepsilon, \mathbf{z} \rangle| \leq \|\varepsilon\| \|\mathbf{z}\| \leq \rho(\Lambda) \|\mathbf{z}\|$ , which is the first claim and holds for every input without any assumption.

For the second claim, the translates  $\{\boldsymbol{\lambda} + \mathcal{V}(\Lambda)\}_{\boldsymbol{\lambda} \in \Lambda}$  tile  $\mathbb{R}^{n_f}$  up to a set of measure zero. Since  $q\mathbb{Z}^{n_f} \subseteq \Lambda$ , the tiling descends to the torus  $\mathbb{R}^{n_f}/q\mathbb{Z}^{n_f}$ , which is covered by exactly  $N_G = q^{n_f}/\det \Lambda$  translates of  $\mathcal{V}(\Lambda)$ , of total volume  $N_G \det \Lambda = q^{n_f}$ , the volume



of the torus. Hence the map  $\mathbf{a} \mapsto \varepsilon$  pushes the uniform distribution on the torus to the uniform distribution on  $\mathcal{V}(\Lambda)$ . Consequently

$$\mathbb{E}\|\varepsilon\|^2 = \frac{1}{\det\Lambda} \int_{\mathcal{V}(\Lambda)} \|\mathbf{x}\|^2 d\mathbf{x} = n_f G(\Lambda) (\det\Lambda)^{2/n_f}$$

by the definition of the normalized second moment. If  $\mathbf{z}$  is independent of  $\mathbf{a}$  with independent entries of mean zero and variance  $\sigma^2$ , then conditionally on  $\varepsilon$  we have  $\mathbb{E}_{\mathbf{z}}[\langle \varepsilon, \mathbf{z} \rangle^2] = \sigma^2 \|\varepsilon\|^2$ , and averaging over  $\varepsilon$  gives  $\sigma^2 n_f G(\Lambda) (\det\Lambda)^{2/n_f}$ . ■

## D. Proof of Theorem 5

We use two facts about the framework of [19]. Write  $f_{L,1/s}(\mathbf{x}) = \rho_{1/s}(\mathbf{x} + L) / \rho_{1/s}(L)$  for the normalized Gaussian mass of a lattice  $L$  at  $\mathbf{x}$  and  $g_W(\mathbf{x}) = \frac{1}{N} \sum_j \cos(2\pi \langle \mathbf{w}_j, \mathbf{x} \rangle / q)$  for the empirical score of a list  $W$ .

*Fact 1.* If  $\mathbf{w}_1, \dots, \mathbf{w}_N$  are drawn independently from  $D_{L_q^\perp(\mathbf{A}_d), q_s}$ , then with probability at least  $1 - q^m 2^{-\Omega(N\delta^2)}$  over that draw,  $|g_W(\mathbf{x}) - f_{L_q(\mathbf{A}_d), 1/s}(\mathbf{x})| \leq \delta$  simultaneously for all  $\mathbf{x} \in \mathbb{Z}_q^m$ . This is the pointwise approximation lemma of [6] combined with Hoeffding's inequality and a union bound over the  $q^m$  points, exactly as in [19].

*Fact 2.* For every lattice  $L \subset \mathbb{R}^m$  and every  $\mathbf{x}$ ,  $f_{L,1/s}(\mathbf{x}) \geq \rho_{1/s}(\|\mathbf{x}\|)$ , and if  $\text{dist}(\mathbf{x}, L) \geq \tau$  with  $\tau = s^{-1} \sqrt{m/2\pi}$  then  $f_{L,1/s}(\mathbf{x}) \leq \rho_{1/s}(\text{dist}(\mathbf{x}, L) - \tau)$ . These are the two bounds on the Gaussian mass used in [19], the second based on Banaszczyk's tail inequality [33].

Condition on the event of Fact 1. Let  $\tilde{\mathbf{s}}$  be a candidate whose coset modulo  $\Lambda_o$  contains a representative of norm at most  $B$ , and take  $\tilde{\mathbf{s}}$  to be that representative. Put  $\Delta = \mathbf{s}_g - \tilde{\mathbf{s}}$ , so  $\|\Delta\| \leq 2B$ . Since  $\mathbf{w}_j \in L_q^\perp(\mathbf{A}_d)$  we have  $\langle \mathbf{w}_j, \mathbf{A}_d \mathbf{s}_d \rangle \equiv 0 \pmod{q}$ , hence with  $\mathbf{a}_j = \mathbf{A}_g^\top \mathbf{w}_j$ ,  $\lambda_j = Q_\Lambda(\mathbf{a}_j)$  and  $\varepsilon_j = \mathbf{a}_j - \lambda_j$ ,  
 $\langle \mathbf{w}_j, \mathbf{b} \rangle - \langle \lambda_j, \tilde{\mathbf{s}} \rangle \equiv \langle \mathbf{a}_j, \mathbf{s}_g \rangle + \langle \mathbf{w}_j, \mathbf{e} \rangle - \langle \lambda_j, \tilde{\mathbf{s}} \rangle$   
 $\equiv \langle \lambda_j, \Delta \rangle + \langle \varepsilon_j, \mathbf{s}_g \rangle + \langle \mathbf{w}_j, \mathbf{e} \rangle$   
 $\equiv \langle \mathbf{w}_j, \mathbf{e} + \mathbf{A}_g \Delta \rangle + \langle \varepsilon_j, \tilde{\mathbf{s}} \rangle \pmod{q}$ ,

the second line by splitting  $\mathbf{a}_j = \lambda_j + \varepsilon_j$  in the term  $\langle \mathbf{a}_j, \mathbf{s}_g \rangle$  and the third by splitting it in the term  $\langle \lambda_j, \tilde{\mathbf{s}} \rangle$  instead and using  $\langle \mathbf{a}_j, \Delta \rangle = \langle \mathbf{w}_j, \mathbf{A}_g \Delta \rangle$ .

*Correct coset.* Suppose  $\tilde{\mathbf{s}} \equiv \mathbf{s}_g \pmod{\Lambda_o}$ . Then  $\Delta \in \Lambda_o = q\Lambda^*$ , so  $\langle \lambda_j, \Delta \rangle \in q\mathbb{Z}$  for every  $\lambda_j \in \Lambda$  and the second line above reduces to  $\langle \varepsilon_j, \mathbf{s}_g \rangle + \langle \mathbf{w}_j, \mathbf{e} \rangle$ . Since  $|\cos(\alpha + \beta) - \cos\alpha| \leq |\beta|$  for all real  $\alpha, \beta$ , and since  $|2\pi \langle \varepsilon_j, \mathbf{s}_g \rangle / q| \leq 2\pi \rho(\Lambda) B / q$  by Lemma 4,

$$\frac{1}{N} S(\tilde{\mathbf{s}}) \geq g_W(\mathbf{e}) - \frac{2\pi \rho(\Lambda) B}{q} \geq \rho_{1/s}(\|\mathbf{e}\|) - \delta - \frac{2\pi \rho(\Lambda) B}{q},$$

using Fact 1 and then the first half of Fact 2. Because  $\|\mathbf{s}_g\| \leq B$  and  $\lambda_1(\Lambda_o) > 2B$ , the correct coset contains exactly one representative of norm at most  $B$ , which is  $\mathbf{s}_g$  itself: any other such representative  $\tilde{\mathbf{s}}$  would give  $\mathbf{s}_g - \tilde{\mathbf{s}} \in \Lambda_o \setminus \{0\}$  of norm at most  $2B < \lambda_1(\Lambda_o)$ , which is impossible.

*Wrong coset.* Suppose  $\tilde{\mathbf{s}} \not\equiv \mathbf{s}_g \pmod{\Lambda_o}$ , so  $\Delta \neq 0$ . From  $\|\Delta\| \leq 2B < q$  we get  $\Delta \notin q\mathbb{Z}^{n_f}$ , hence  $\Delta \not\equiv 0 \pmod{q}$ . Because  $\mathbf{A}$  has full rank over  $\mathbb{Z}_q$  and  $m \geq n$ , the vector  $\mathbf{A}_g \Delta$  does not lie in  $L_q(\mathbf{A}_d)$ : otherwise  $\mathbf{A}_g \Delta \equiv \mathbf{A}_d \mathbf{y} \pmod{q}$  for some  $\mathbf{y}$ , so the nonzero vector  $(\Delta, -\mathbf{y})$  of  $\mathbb{Z}_q^n$  would be in the kernel of  $\mathbf{A}$ . Consequently  $\mathbf{A}_g \Delta - \mathbf{v}$  is a nonzero vector of  $L_q(\mathbf{A})$  for every  $\mathbf{v} \in L_q(\mathbf{A}_d)$ , whence  $\text{dist}(\mathbf{A}_g \Delta, L_q(\mathbf{A}_d)) \geq \lambda_1(L_q(\mathbf{A}))$  and, by the triangle inequality,

$$\text{dist}(\mathbf{e} + \mathbf{A}_g \Delta, L_q(\mathbf{A}_d)) \geq \lambda_1(L_q(\mathbf{A})) - \|\mathbf{e}\| \geq \tau.$$

Using the third line of the display above,  $\|\tilde{\mathbf{s}}\| \leq B$  and Lemma 4 once more,

$$\begin{aligned} \frac{1}{N} S(\tilde{\mathbf{s}}) &\leq g_W(\mathbf{e} + \mathbf{A}_g \Delta) + \frac{2\pi \rho(\Lambda) B}{q} \\ &\leq \rho_{1/s}(\lambda_1(L_q(\mathbf{A})) - \|\mathbf{e}\| - \tau) + \delta + \frac{2\pi \rho(\Lambda) B}{q}, \end{aligned}$$

by Fact 1 and the second half of Fact 2,  $\rho_{1/s}$  being decreasing on  $[0, \infty)$ .

Subtracting the two bounds, the score of the correct coset exceeds the score of every wrong admissible coset as soon as

$$\rho_{1/s}(\|\mathbf{e}\|) - \rho_{1/s}(\lambda_1(L_q(\mathbf{A})) - \|\mathbf{e}\| - \tau) > 2\delta + \frac{4\pi \rho(\Lambda) B}{q},$$

which is the hypothesis (3). Since the correct coset is admissible, because  $\|\mathbf{s}_g\| \leq B$ , the maximum in line 9 of Algorithm 1 is attained at it, and the shortest representative returned is  $\mathbf{s}_g$  by the uniqueness established above.

For the running time, the sampling step is  $N$  calls to a discrete Gaussian sampler, the loop performs  $N$  quantizations and  $N$  inner products, each of cost  $\text{poly}(m, n)$ , the transform costs  $O(N_G \log N_G)$  arithmetic operations by Lemma 2, and the admissibility test inspects each of the  $N_G$  cosets once. The space is one accumulator of  $N_G$  complex words together with the instance, that is  $O(N_G + mn)$ . The failure probability is that of Fact 1. ■

## E. Proof of Theorem 6

*Quantizer lattice.* For a block  $b$  and a coordinate  $i$  of that block,  $q\mathbf{e}_i = \frac{q}{c_b m_b} \cdot (c_b m_b \mathbf{e}_i)$  and  $c_b m_b \mathbf{e}_i = m_b (c_b \mathbf{e}_i) \in \Lambda_b$ , because  $c_b \mathbb{Z}^{n_b} \subseteq \Lambda_b$  and  $\Lambda_b$  is a group and  $m_b$  an integer. Hence  $q\mathbb{Z}^{n_f} \subseteq \Lambda$ , and  $\Lambda$  has full rank as a direct sum of full-rank lattices.

*Table size.*  $\det\Lambda = \prod_b (q/(c_b m_b))^{n_b} \det\Lambda_b$ , so

$$N_G = \frac{q^{n_f}}{\det\Lambda} = \prod_b \frac{(c_b m_b)^{n_b}}{\det\Lambda_b},$$

and taking base-two logarithms gives the stated formula. Each  $c_b m_b$  is a power of two and each  $\det\Lambda_b$  is a power of two for the lattices of Table I, so  $\log_2 N_G$  is an integer. Every integer in the admissible range is attained because the family contains the block  $\Lambda_b = \mathbb{Z}$  with  $c_b = 1$  and  $\det\Lambda_b = 1$ , which contributes exactly  $\log_2 m_b$  bits and therefore any nonnegative integer, so a design of  $n_f - 1$  coordinates can be completed to the required total by one such block.

*Index group.* Multiplication by  $c_b m_b / q$  is a group isomorphism from  $\frac{q}{c_b m_b} \Lambda_b / q\mathbb{Z}^{n_b}$  onto  $\Lambda_b / (c_b m_b) \mathbb{Z}^{n_b}$ , and the index group of a direct sum is the direct sum of the index groups, which gives  $G_\Lambda \cong \bigoplus_b \Lambda_b / (c_b m_b) \mathbb{Z}^{n_b}$ . Each summand has order  $(c_b m_b)^{n_b} / \det\Lambda_b$ , a power of two, and exponent dividing  $c_b m_b$ , again a power of two, so  $G_\Lambda$  is an abelian two-group and by the structure theorem is isomorphic to  $\prod_j \mathbb{Z}_{2^{a_j}}$  with  $\sum_j a_j = \log_2 N_G$ . Concretely, let  $\mathbf{B}_b$  be a basis matrix of  $\Lambda_b$  whose columns are basis vectors, so that a point of  $\Lambda_b$  is  $\mathbf{B}_b \mathbf{u}$  with  $\mathbf{u} \in \mathbb{Z}^{n_b}$ . The sublattice  $(c_b m_b) \mathbb{Z}^{n_b}$  corresponds in these coordinates to the row space of  $\mathbf{K}_b = c_b m_b \mathbf{B}_b^{-1}$ , which is an integer matrix precisely because  $(c_b m_b) \mathbb{Z}^{n_b} \subseteq \Lambda_b$ . Writing the Smith normal form  $\mathbf{U} \mathbf{K}_b \mathbf{V} = \mathbf{D} = \text{diag}(d_1, \dots, d_{n_b})$  with  $\mathbf{U}, \mathbf{V}$  unimodular, the map  $\mathbf{u} \mapsto \mathbf{U} \mathbf{u} \pmod{\mathbf{D}}$  is an isomorphism  $\mathbb{Z}^{n_b} / \mathbf{K}_b \mathbb{Z}^{n_b} \rightarrow \prod_j \mathbb{Z}_{d_j}$ , and from  $\mathbf{B}_b = c_b m_b \mathbf{K}_b^{-1}$  one gets

$$\frac{\langle \mathbf{B}_b \mathbf{u}, \tilde{\mathbf{s}} \rangle}{c_b m_b} = \mathbf{u}^\top \mathbf{U}^\top \mathbf{D}^{-1} \mathbf{V}^\top \tilde{\mathbf{s}} = \sum_j \frac{(\mathbf{U} \mathbf{u})_j (\mathbf{V}^\top \tilde{\mathbf{s}})_j}{d_j} \pmod{1},$$

so the pairing of Lemma 2 becomes the standard pairing of  $\prod_j \mathbb{Z}_{d_j}$  with itself once a point is indexed by  $\mathbf{U} \mathbf{u} \pmod{\mathbf{D}}$  and a candidate by  $\mathbf{V}^\top \tilde{\mathbf{s}} \pmod{\mathbf{D}}$ .

*Transform.* A Fourier transform over  $\prod_j \mathbb{Z}_{d_j}$  is the tensor product of the one-dimensional transforms, so it can be evaluated by applying a length- $d_j$  transform along each axis in turn. All  $d_j$  are powers of two, so each axis is handled by the radix-two Cooley–Tukey algorithm; the total number of complex additions is  $N_G \sum_j \log_2 d_j = N_G \log_2 N_G$  and the number of twiddle multiplications is at most half that, every twiddle being a root of unity of order dividing  $\max_j d_j$ , a power of two. When every  $d_j$  equals two the length-two transform is the butterfly  $(a, b) \mapsto (a + b, a - b)$  and no multiplication occurs at all.

*Candidate lattice.* If  $\Lambda_b^* = c_b^{-1} \Lambda_b$  then

$$\Lambda_0 = q\Lambda^* = q \bigoplus_b \frac{c_b m_b}{q} \Lambda_b^* = \bigoplus_b c_b m_b \cdot c_b^{-1} \Lambda_b = \bigoplus_b m_b \Lambda_b,$$

which is integral, and the minimum distance of a direct sum is the least of the minimum distances of the summands, that is  $\min_b m_b \lambda_1(\Lambda_b)$ . ■

### F. Proof of Theorem 7

The point  $Q_{\Lambda'}(\mathbf{a})$  lies in  $\Lambda'$  and  $Q_{\Lambda}(Q_{\Lambda'}(\mathbf{a}))$  lies in  $\Lambda$ , so it is one of the points over which the minimum defining  $Q_{\Lambda}(\mathbf{a})$  is taken. Since  $Q_{\Lambda}(\mathbf{a})$  realizes that minimum,

$$\|\mathbf{a} - Q_{\Lambda}(\mathbf{a})\| = \min_{\lambda \in \Lambda} \|\mathbf{a} - \lambda\| \leq \|\mathbf{a} - Q_{\Lambda}(Q_{\Lambda'}(\mathbf{a}))\|,$$

which holds for every input and needs no assumption. The inequality is strict on a set of positive measure whenever  $\Lambda \subsetneq \Lambda'$ , because the staged map is constant on each cell of  $\Lambda'$  while  $Q_{\Lambda}$  is not.

For the quantitative part, write  $\varepsilon_1 = \mathbf{a} - Q_{\Lambda'}(\mathbf{a})$  and  $\varepsilon_2 = Q_{\Lambda'}(\mathbf{a}) - Q_{\Lambda}(Q_{\Lambda'}(\mathbf{a}))$ , so the staged residual is  $\varepsilon_1 + \varepsilon_2$ . By the argument in the proof of Lemma 4 applied to  $\Lambda'$ , the variable  $\varepsilon_1$  is uniform on  $\mathcal{V}(\Lambda')$  when  $\mathbf{a}$  is uniform modulo  $q\mathbb{Z}^{n_f}$ , so  $\mathbb{E}\|\varepsilon_1\|^2 = n_f G(\Lambda') (\det \Lambda')^{2/n_f}$ . Under the stated hypothesis that the points of  $\Lambda'$  are equidistributed in the cells of  $\Lambda$ , the variable  $\varepsilon_2$  is uniform on  $\mathcal{V}(\Lambda)$  and is a function of  $Q_{\Lambda'}(\mathbf{a})$  alone, hence independent of  $\varepsilon_1$ , which is a function of the position of  $\mathbf{a}$  inside its  $\Lambda'$  cell. Both have mean zero because the cells are symmetric about the origin, so

$$\begin{aligned} \mathbb{E}\|\varepsilon_1 + \varepsilon_2\|^2 &= \mathbb{E}\|\varepsilon_1\|^2 + \mathbb{E}\|\varepsilon_2\|^2 \\ &= n_f G(\Lambda') (\det \Lambda')^{2/n_f} \\ &\quad + n_f G(\Lambda) (\det \Lambda)^{2/n_f}, \end{aligned}$$

while the one-step residual has mean square  $n_f G(\Lambda) (\det \Lambda)^{2/n_f}$  by Lemma 4. The excess is therefore exactly the distortion of the first stage. For  $\Lambda' = (q/p)\mathbb{Z}^{n_f}$  one has  $G(\Lambda') = 1/12$  and  $(\det \Lambda')^{2/n_f} = (q/p)^2$ , so the excess is  $n_f q^2 / (12p^2)$ . ■

### G. Proof of Lemma 8

Let  $V_b = \det(\alpha_b \Lambda_b)$  be the volume of block  $b$ , so that the total volume is  $V = \prod_b V_b$  and, by Lemma 4 applied blockwise, the distortion is  $D = \sum_b n_b G(\Lambda_b) V_b^{2/n_b}$ . Substituting  $x_b = \ln V_b$  turns the constraint into  $\sum_b x_b = \ln V$  and the objective into  $D = \sum_b n_b G(\Lambda_b) e^{2x_b/n_b}$ , a sum of convex functions of  $x_b$ , hence convex; the constraint is affine, so a stationary point of the Lagrangian is the global minimum. Setting  $\partial/\partial x_b$  of  $D - \mu \sum_b x_b$  to zero gives  $2G(\Lambda_b) e^{2x_b/n_b} = \mu$ , that is  $G(\Lambda_b) V_b^{2/n_b} = \mu/2$  for every  $b$ , which is the equalization claim. Then  $D = \sum_b n_b \mu/2 = n_f \mu/2$ , and

$$\begin{aligned} V &= \prod_b V_b = \prod_b \left( \frac{\mu}{2G(\Lambda_b)} \right)^{n_b/2}, \\ V^{2/n_f} &= \frac{\mu}{2} \prod_b G(\Lambda_b)^{-n_b/n_f}. \end{aligned}$$

Combining the last two displays,  $D/n_f = V^{2/n_f} \prod_b G(\Lambda_b)^{n_b/n_f}$ , and comparing with the definition  $D = n_f G(\Lambda) V^{2/n_f}$  gives  $G(\Lambda) = \prod_b G(\Lambda_b)^{n_b/n_f}$ . ■

### H. Proof of Theorem 9

By definition  $\mathbb{E}[F_{\mathbf{w}}(\mathbf{s}_f)] = \sum_j w_j \mathbb{E}[\cos(2\pi(\nu_j + \langle \varepsilon_j, \mathbf{s}_f \rangle)/q)] = \sum_j w_j \tau_j = \langle \mathbf{w}, \boldsymbol{\tau} \rangle$ , the expectation being over the error and the parts of the secret that are not being tested. For a candidate drawn uniformly among the  $N_G$  cosets, Lemma 3 gives mean  $\text{Re}h(0)$  and second moment  $\frac{1}{2} \sum_j w_j^2$  under the stated conditions on the indices, so the variance is  $\frac{1}{2} \|\mathbf{w}\|_2^2$  up to the square of  $\text{Re}h(0)$ , which vanishes when no sample falls in the zero coset. Hence

$$\Delta(\mathbf{w}) = \frac{\langle \mathbf{w}, \boldsymbol{\tau} \rangle}{\sqrt{\frac{1}{2} \|\mathbf{w}\|_2^2}} = \sqrt{2} \frac{\langle \mathbf{w}, \boldsymbol{\tau} \rangle}{\|\mathbf{w}\|_2}.$$

By the Cauchy–Schwarz inequality  $\langle \mathbf{w}, \boldsymbol{\tau} \rangle \leq \|\mathbf{w}\|_2 \|\boldsymbol{\tau}\|_2$ , with equality if and only if  $\mathbf{w}$  and  $\boldsymbol{\tau}$  are proportional, so  $\Delta(\mathbf{w}) \leq \sqrt{2} \|\boldsymbol{\tau}\|_2$  with equality exactly at  $\mathbf{w} \propto \boldsymbol{\tau}$ . Since  $\Delta$  is invariant under positive scaling of  $\mathbf{w}$ , the maximizer is unique up to that scaling.

For uniform weights  $\mathbf{w} = \mathbf{1}$  one gets  $\Delta(\mathbf{1}) = \sqrt{2} \|\boldsymbol{\tau}\|_1 / \sqrt{N}$ , so the ratio of the two deflections is  $\sqrt{N} \|\boldsymbol{\tau}\|_2 / \|\boldsymbol{\tau}\|_1$ , which is at least one by Cauchy–Schwarz applied to  $\|\boldsymbol{\tau}\|_1 = \langle \mathbf{1}, \boldsymbol{\tau} \rangle \leq \sqrt{N} \|\boldsymbol{\tau}\|_2$ , with equality if and only if  $\boldsymbol{\tau}$  is proportional to  $\mathbf{1}$ , that is all  $\tau_j$  are equal. Finally, if the  $N$  samples are drawn from one distribution then  $\|\boldsymbol{\tau}\|_2^2$  concentrates around  $N\mathbb{E}[\tau^2]$  and  $\|\boldsymbol{\tau}\|_1$  around  $N\mathbb{E}[\tau]$ , so the optimal and the uniform deflections are  $\sqrt{2N\mathbb{E}[\tau^2]}$  and  $\sqrt{2N\mathbb{E}[\tau]}$ ; equating either to a fixed target and solving for  $N$  shows that the sample count needed by the uniform rule exceeds the one needed by the optimal rule by the factor  $\mathbb{E}[\tau^2]/\mathbb{E}[\tau]^2$ , which is the square of the gain. ■

### I. Proof of Corollary 10

Appending a sample replaces  $\boldsymbol{\tau} \in \mathbb{R}_{\geq 0}^N$  by  $(\boldsymbol{\tau}, \tau_{N+1}) \in \mathbb{R}_{\geq 0}^{N+1}$ , so  $\|\boldsymbol{\tau}\|_2^2$  increases by  $\tau_{N+1}^2 \geq 0$ . By Theorem 9 the optimal deflection is  $\sqrt{2} \|\boldsymbol{\tau}\|_2$ , a nondecreasing function of that quantity, so it does not decrease. The same is not true of the uniform rule, whose deflection  $\sqrt{2} \|\boldsymbol{\tau}\|_1 / \sqrt{N}$  decreases as soon as  $\tau_{N+1}$  falls below the current average  $\|\boldsymbol{\tau}\|_1 / N$ . ■

### J. Proof of Proposition 11

*Scalar modulus switching.* Each coordinate of  $\mathbf{a}$  is multiplied by  $p/q$ , rounded and multiplied back, which is a constant number of operations per coordinate, hence  $\Theta(n_f)$  in total, and only the current vector is held, hence  $O(n_f)$  words.

*Nested lattice quantizer.* The blocks are processed independently. A block  $\mathbb{Z}$  costs one rounding. A block  $D_k$  costs one rounding per coordinate plus one parity correction, that is  $\Theta(k)$  operations [26]. A block  $\Lambda_{E_8}$  is decoded by the two-coset algorithm of [26], which performs two  $D_8$  decodings and one comparison, 104 operations. A block  $\Lambda_{24}$  is decoded by the algorithm of [27], which performs 331 operations. Summing over  $\lceil n_f/n_b \rceil$  blocks gives  $\Theta(n_f)$  operations with the stated constants, and each decoder stores only the current block, hence  $O(n_f)$  words.

*Polar source code over  $\mathbb{Z}_q$ .* The successive cancellation recursion on a block of length  $n'$  visits  $n' \log_2 n'$  nodes. At a node of the first kind the likelihood vector of a variable is the cyclic convolution of two length- $q$  vectors, computed by three transforms of length  $q$  at cost  $\Theta(\mu(q))$ ; at a node of the second kind it is a shifted pointwise product at cost  $\Theta(q)$ . The total is  $\Theta(n' \log n' \cdot \mu(q))$ . The recursion keeps one length- $q$  likelihood vector per position at each of the  $\log_2 n'$  levels that are alive, which is  $\Theta(n' q)$  words.

*Staged switching and coding.* A rounding pass at cost  $\Theta(n_f)$  is followed by the same recursion over  $\mathbb{Z}_p$ , giving  $\Theta(n_f + n' \log n' \cdot \mu(p))$  operations and  $\Theta(n' p)$  words.

*Product of random codes.* Each of the  $n_f/n_0$  blocks is decoded by exhaustive search over its  $q^{k_0}$  codewords, each comparison costing  $\Theta(n_0)$ , hence  $\Theta((n_f/n_0) q^{k_0} n_0)$  operations, and the codebook of one block occupies  $\Theta(q^{k_0} n_0)$  words.

*Transforms.* By Lemma 2 the score vector is a Fourier transform over a finite abelian group of order  $N_G$ , which factors as a multidimensional transform. For the two-group of Theorem 6 all radices are two, so the cost is  $\Theta(N_G \log_2 N_G)$  additions and no length- $q$  subtransform occurs. For  $\mathbb{Z}_q^{k_f}$  the transform is  $k_f$  passes of  $q^{k_f-1}$  transforms of length  $q$ , that is  $k_f q^{k_f-1} \mu(q)$  operations; writing  $k_f = \log_q N_G$  this equals  $\Theta(N_G \log_2 N_G)$  with a constant proportional to  $\mu(q)/(q \log_2 q)$ . In both cases the accumulator holds  $\Theta(N_G)$  complex words. ■

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